Autonomous Driving Project – Team Task Division

#	Task	Short Summary	Ali Abdelkhalek	Mohamed Awis	Yassin Elhalawany
1	Git & Project Setup	Create Git repo and organize project	**Created the Task devision File**	**Created the Git repo**	**Organized the files for the first meeting**
2	System Engineering	Develop SysML diagrams and requirements	**Created Requirements diagaram and Internal Block Diagram (IBD)**	**Created Use case diagaram, Activity diagram and Package diagram**	**Created Sequence diagaram and Block Definition Diagram (BDD)**
3	Prototype Simulation	Develop first Tinkercad prototype		**Created TinkerCad simulation circuit (with coding for detecting the line only) and electrical circuit**	
4	Line Detection	Vehicle follows lane autonomously	**Performed cable management and wire routing, organized and assembled components, optimized the code to suit the available hardware conditions, implemented black line detection and tracking, calibrated IR sensors, integrated system components, Speed Optimization and Performance Tuning for Improved Line-Tracking Accuracy and Stability and conducted real-world testing and performance evaluation of the car.**		
5	**Milestone 1**	**The Vehicle Should follows the line**	**Achived and the car was able to follow the black line smoothly without any issues**		
6	Routing Functions	Oval and figure-eight routing	**Optimized differential motor control, turning behavior, speed parameters, and sensor response to achieve smoother movement, more reliable curve tracking, and stable autonomous operation.**		
7	Parking Function,180° Turn and Line Reacquisition (BONUS)	Autonomous parking capability	**Autonomous parking was excluded from the final implementation. Instead, after detecting the second obstacle, the vehicle performs a 180° turn, reacquires the line, and continues autonomous line following.**		
8	Obstacle Detection	Detect and avoid obstacles	**Integrated and calibrated the ultrasonic sensors, developed the obstacle-detection and state-based avoidance logic, optimized maneuver timing and motor speeds, and tested the complete sequence of obstacle detection, avoidance, line reacquisition, and continued line following.**		
9	Milestone 2	Vehicle detects obstacles successfully	**Successfully validated the integrated system: the vehicle follows the line, detects and avoids the first obstacle, reacquires the line, responds to the second obstacle with a 180° turn, and continues autonomous line following.**		